



भारतीय प्रौद्योगिकी संस्थान दिल्ली
Indian Institute of Technology Delhi

Concrete Strength Prediction and RCC Suitability Classification

Course Project: AIL7024

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Problem Statement & Motivation

The Challenge



Predicting concrete's compressive strength is time-consuming and costly, requiring weeks of lab testing to cast, cure, and test samples.

1. Why Predict Strength? (Regression)



To get early feedback on new concrete mix designs.



To understand how different ingredients (cement, water, etc.) impact the final strength.

2. Why Classify Suitability? (Classification)



To make a direct, actionable decision: "Is this mix suitable for Reinforced Cement Concrete (RCC)?"

Engineering Standard: IS 456:2000

A mix is suitable if its strength is ≥ 20 MPa at an age of ≥ 28 days.

Dataset Summary



Source: UCI Machine Learning Repository
(Dataset ID: 165)



Instances: 1030 samples with no missing values



Features: 8 input variables describing the concrete mix composition and age.



Target (Regression): Compressive Strength (in MPa)

Feature	Description
Cement	Cement content (kg/m ³)
Blast Furnace Slag	Slag content (kg/m ³)
Fly Ash	Fly ash content (kg/m ³)
Water	Water content (kg/m ³)
Superplasticizer	Superplasticizer content (kg/m ³)
Coarse Aggregate	Coarse aggregate content (kg/m ³)
Fine Aggregate	Fine aggregate content (kg/m ³)
Age	Age of concrete (days)

Feature Engineering

Rationale: Raw component values are not enough. Ratios and interactions based on material science provide deeper insights into the chemical processes governing strength.



Totals

Total Binder = Cement + Blast Furnace Slag + Fly Ash

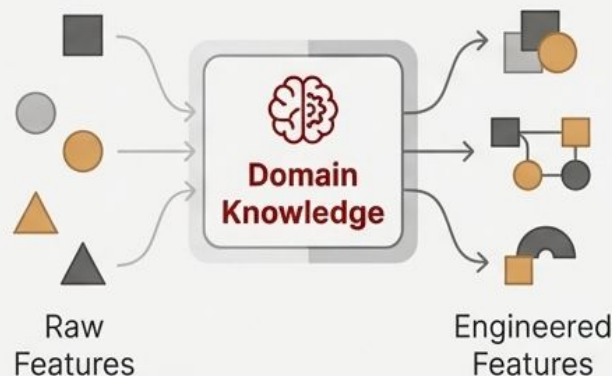
Total Aggregate = Coarse Aggregate + Fine Aggregate



Critical Ratios

$$\text{Water/Cement Ratio} = \frac{\text{Water}}{\text{Cement} + \epsilon}$$
$$\text{Binder-Aggregate Ratio} = \frac{\text{Total Binder}}{\text{Total Aggregate} + \epsilon}$$
$$\text{SP_per_Binder} = \frac{\text{Superplasticizer}}{\text{Total Binder} + \epsilon}$$

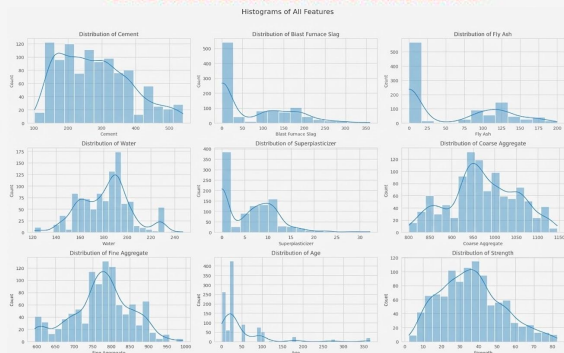

Age Transformation

$$\text{Age_Log} = \log(\text{Age} + 1)$$


EDA Highlights

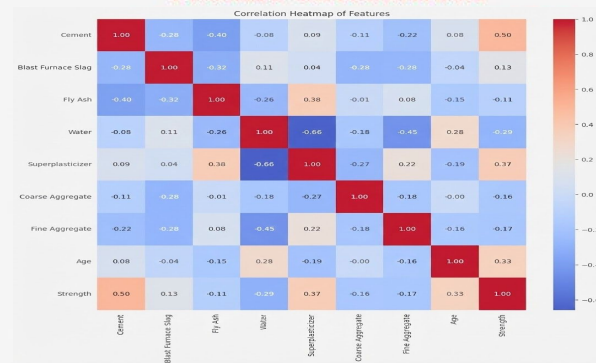
Insights from Visualizing the Data in Inter Regular

Distribution of Features



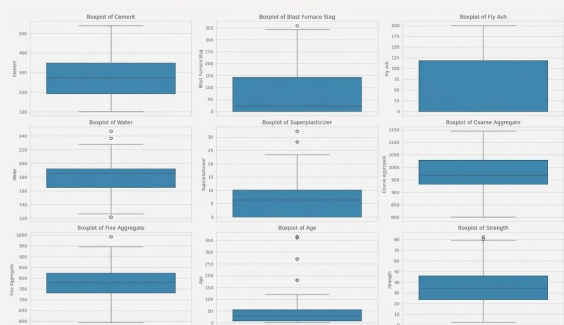
*Age and Supersplasticizer are heavily skewed.

Feature Correlations



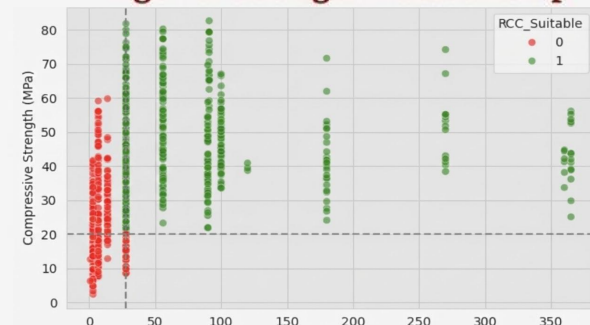
*Cement has the strongest positive correlation with Strength (0.50).

Feature Outliers



*Age shows significant outliers, suggesting non-linear effects.

Age vs. Strength Relationship



*Strength increases with Age, but the relationship plateaus. The RCC suitability thresholds (dashed lines) are clearly visible.

Regression Modeling

Objective: Predict the exact compressive strength (MPa).

Models Implemented

From Scratch:

Built to understand their mechanics.

Normal Equation: $\theta = (X^T X)^{-1} X^T y$

Ridge Regression: $\theta = (X^T X + \alpha I)^{-1} X^T y$

Lasso (Coordinate Descent): $\theta_j \leftarrow S(\rho_j, \lambda_n) / z_j$

Benchmark Models:

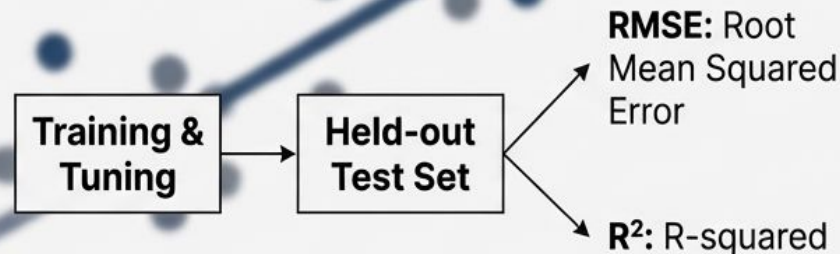
Advanced library models for top performance.

Support Vector Regression (SVR)

XGBoost

LightGBM

Evaluation



All models were trained and tuned, then evaluated on a held-out test set using Root Mean Squared Error (RMSE) and R-squared (R^2).

Best Regression Model & Results

Best Performing Model: LightGBM

Achieved the lowest error and highest explained variance after hyperparameter tuning.

Test RMSE

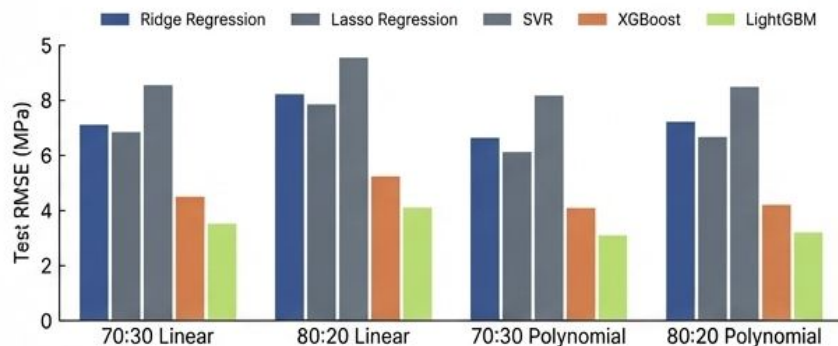
Test RMSE: 4.33 MPa

Test R^2

Test R^2 : 0.9308

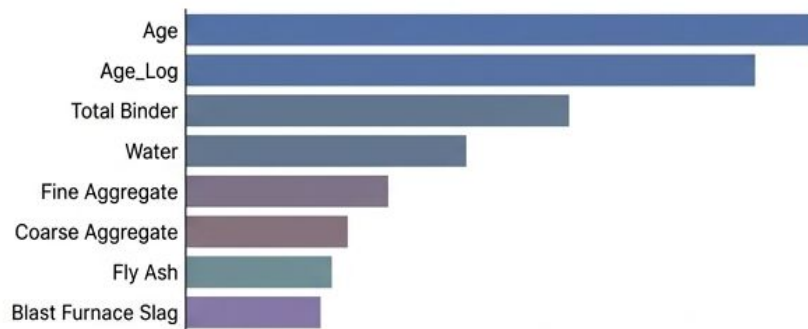
* This means our model's predictions are, on average, only 4.33 MPa off the true value, and it explains over 93% of the variance in concrete strength.

Model Performance Comparison (After Tuning)



*LightGBM and XGBoost clearly outperform linear models and SVR.

Feature Importance (from Gradient Boosting)



*Age, Log-Transformed Age, and Total Binder are the most predictive features.

Regression Results: Performance After Hyperparameter Tuning

Dataset	Model	CV RMSE	Test MSE	Test RMSE	Test R2
70:30 Linear	Ridge Regression	10.3890	109.0886	10.4445	0.5968
80:20 Linear	Ridge Regression	10.5191	95.9651	9.7962	0.6276
70:30 Polynomial	Ridge Regression	6.4835	47.1203	6.8644	0.8259
80:20 Polynomial	Ridge Regression	6.6524	41.7222	6.4593	0.8381
70:30 Linear	Lasso Regression	10.4122	108.7401	10.4279	0.5981
80:20 Linear	Lasso Regression	10.5190	95.9485	9.7953	0.6276
70:30 Polynomial	Lasso Regression	6.5174	47.0018	6.8558	0.8263
80:20 Polynomial	Lasso Regression	6.6189	41.7097	6.4583	0.8381
70:30 Linear	SVR	7.3283	57.9170	7.6103	0.7859
80:20 Linear	SVR	7.2071	51.1047	7.1488	0.8017
70:30 Polynomial	SVR	5.7338	31.0164	5.5692	0.8854
80:20 Polynomial	SVR	5.6130	28.5399	5.3423	0.8892
70:30 Linear	XGBoost	4.9295	21.9325	4.6832	0.9189
80:20 Linear	XGBoost	4.6661	19.3635	4.4004	0.9249
70:30 Polynomial	XGBoost	4.9447	20.7670	4.5571	0.9232
70:30 Polynomial	XGBoost	4.9447	22.7670	4.5571	0.9118
80:20 Polynomial	XGBoost	4.7358	22.7184	4.7664	0.9118
80:20 Linear	LightGBM	4.4472	21.0364	4.5865	0.9184
70:30 Polynomial	LightGBM	4.7843	19.5419	4.4206	0.9278
80:20 Polynomial	LightGBM	4.6032	21.8454	4.6739	0.9152

Key Findings:

- Non-linear models (SVR, XGBoost, LightGBM) significantly outperformed linear models.
- Including polynomial features consistently improved performance for Ridge, Lasso, and SVR.
- Tree-based ensembles (XGBoost, LightGBM) most effectively captured complex interactions, achieving the lowest errors.

Classification Pipeline

New Objective: Is a concrete mix suitable for RCC structures?

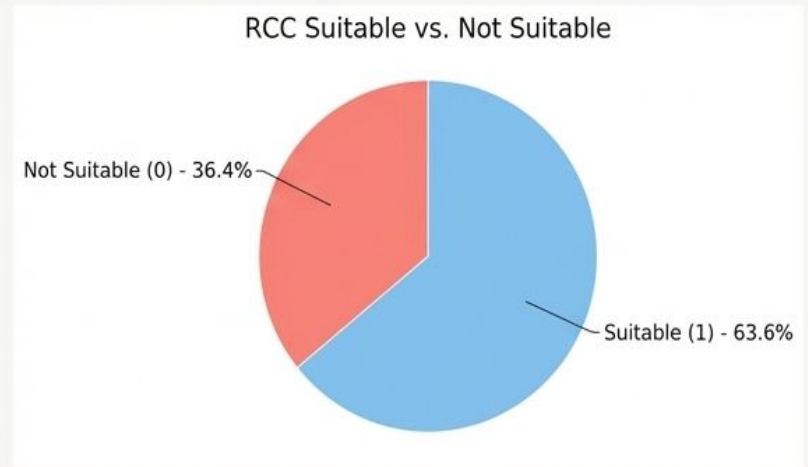
Target Definition (Based on IS 456:2000)

$$\text{RCC Suitable} = \begin{cases} 1, \checkmark & \text{if Strength} \geq 20 \text{ MPa and Age} \geq 28 \text{ days} \\ 0, \times & \text{otherwise} \end{cases}$$

Pipeline

1. Create **binary** RCC_Suitable label.
2. Use **engineered features** from the previous stage ('Age_Log', 'Water/Cement Ratio', etc.).
3. Handle class imbalance in the training data using **SMOTE**.
4. Train and tune multiple classifiers (Logistic Regression, SVM, Random Forest, etc.).

Class Distribution



Classification Results: Evaluating RCC Suitability

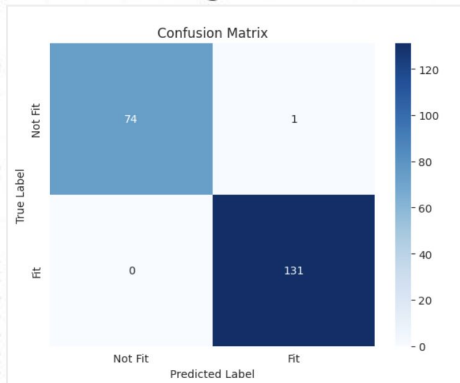
Model	Accuracy	Precision	Recall	F1 Score	AUC Score
LogReg (Sklearn Tuned)	0.9951	0.9924	1.0000	0.9962	0.9991
SVM	0.9903	0.9850	1.0000	0.9924	0.9990
SVM (Tuned)	0.9854	0.9776	1.0000	0.9887	0.9977
Decision Tree	0.9854	0.9848	0.9924	0.9886	0.9828
Gradient Boosting (Tuned)	0.9806	0.9704	1.0000	0.9850	0.9997

Best Model: Logistic Regression (Sklearn Tuned)

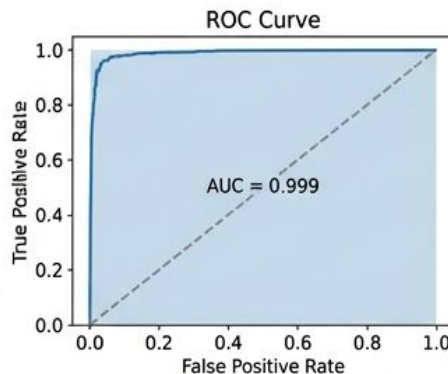
F1 Score: 0.9962

AUC Score: 0.9991

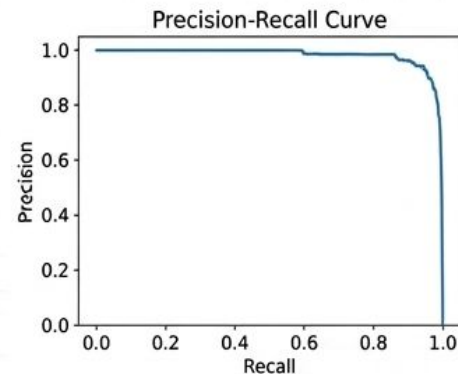
This simple, interpretable model achieved near-perfect performance after tuning, demonstrating the high predictive power of our engineered feature set.



Excellent separation: 0 False Negatives and only 1 False Positive.



AUC of 0.999 shows outstanding discriminative ability.



The curve hugs the top-right corner, indicating high precision and recall.

Deployment Function

Goal: Create a simple, reusable function to assess any new concrete mix.



Input: New Concrete Mix

**(Cement, Water, Age, etc.)*



Step 1: Feature Engineering

Calculate Ratios & Totals (Water/Cement, Total Binder, etc.)



Step 2: Scaling

Apply Pre-fitted StandardScaler (Use statistics from training data)



Step 3: Prediction

Use Trained Logistic Regression Model



Output

1. **Prediction:** 'Fit for RCC' or 'Not Fit'
2. **Probability:** Confidence score (e.g., 99.8%)

Conclusions & Limitations

Conclusions

- ✓ Successfully built a dual pipeline for both **predicting** strength and **classifying** RCC suitability.
- ✓ Domain-specific **feature engineering** was the key driver of high performance in both tasks.
- ✓ **Non-linear models (LightGBM)** excelled at regression, achieving an R^2 of 0.93.
- ✓ A tuned **Logistic Regression** model achieved a near-perfect F1-score of 0.9962 for classification, making it a reliable and interpretable choice for deployment.

Limitations & Future Work

- ⚠ The classification label is rule-derived from the data itself; real-world labels may have more noise.
- ⚠ The dataset, while robust, covers a specific range of concrete designs. The model should be validated on more diverse, real-world field data.
- ⚠ SMOTE generates synthetic data, which may not always represent physically achievable concrete mixes.



Thank You

Questions?

Course Code: AIL7024